

Benchmarking Deepfake Detection Methods

A Comparative Study of Four Representative Architectures Under a Unified Pipeline

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Abstract

Deepfake technology poses an increasingly serious threat to digital media integrity, enabling the synthesis of convincing facial manipulations that are difficult to detect by human observers. This report presents a comparative benchmarking study of four representative deepfake detection architectures—Meso4, Xception, Patch ResNet, and Multiple Attention—evaluated under a unified training and evaluation pipeline. Each model is assessed across a broad suite of detection performance metrics (Accuracy, Balanced Accuracy, F1 Score, ROC-AUC, PR-AUC, and MCC) as well as computational efficiency metrics (FLOPs, parameter count, model size, FPS, and inference latency). Xception achieves the highest detection performance with an ROC-AUC of 0.879 and an MCC of 0.510, while Meso4 attains the fastest inference speed at over 6,000 FPS. The Multiple Attention model provides a compelling intermediate trade-off, approaching Xception closely in accuracy while consuming substantially fewer computational resources. Patch ResNet offers a solid balance between efficiency and detection capability. This work is inspired by and based on the benchmarking methodology proposed by Deng et al. (2024) in “Towards Benchmarking and Evaluating Deepfake Detection.”

1 Introduction

Deepfake technology, driven by advances in generative deep learning, enables the realistic synthesis and replacement of facial identities in images and video. Since its emergence, the quality and accessibility of face manipulation tools have improved dramatically, enabling the creation of highly convincing synthetic media with consumer-grade hardware. The societal implications are significant: deepfakes have been weaponised for political disinformation, non-consensual intimate imagery, financial fraud, and identity theft.

In response, the research community has proposed numerous automated deepfake detection methods. However, a persistent challenge is the absence of a standardised evaluation protocol. Many published results are not directly comparable because models are trained on different datasets, use different preprocessing pipelines, or are evaluated against inconsistent test sets. This makes it difficult to determine whether a reported performance improvement stems from the detection method itself or from more favourable training conditions—a gap identified and addressed by Deng et al. (2024).

This project addresses that gap by implementing four architecturally distinct deepfake detection methods and training each under an identical experimental pipeline on the same dataset split. The evaluation covers both detection quality and computational efficiency, enabling a practical comparison suited to real-world deployment decisions.

1.1 Project Objectives

- Implement four representative deepfake detection architectures spanning the lightweight CNN, deep CNN, patch-based CNN, and attention-based paradigms.
- Train every model under a common preprocessing and training protocol to eliminate confounding variables.
- Evaluate all models on the same held-out test set using a comprehensive set of detection and efficiency metrics.
- Analyse the accuracy-efficiency trade-off and derive actionable recommendations for practical deployment.

1.2 Contributions

- A reproducible unified benchmark comparing Meso4, Xception, Patch ResNet, and Multiple Attention on an identical dataset and pipeline.
- Comprehensive reporting across nine detection metrics and five efficiency metrics, many of which are absent from existing single-model publications.
- A detailed analysis of the accuracy-versus-speed frontier, including ROC curves, Precision-Recall curves, confusion matrices, and a parameter count comparison.
- Practical guidance on model selection across deployment scenarios ranging from edge inference to high-accuracy server-side detection.

2 Related Work

2.1 Deepfake Generation

Contemporary facial manipulation techniques generally fall into three primary categories: autoencoder-driven frameworks (such as FaceSwap and FakeApp), Generative Adversarial Networks (including FaceShifter and FSGAN), and computer graphics-based pipelines utilising geometric facial landmarks. Autoencoder paradigms facilitate identity replacement by mapping distinct faces onto a shared latent space; however, decoded results frequently exhibit limited spatial resolution. GAN-based architectures leverage adversarial training to synthesise high-fidelity alterations that effectively manage environmental lighting and complex occlusions. Graphic-based strategies achieve identity substitution by deforming geometric facial meshes, offering lower photorealism but greater data efficiency. The convergence of these methodologies across benchmark datasets—specifically FaceForensics++, Celeb-DF, DeepFake-TIMIT, DFDC, and ForgeryNet—yields a diverse spectrum of forgery signatures against which detectors are evaluated.

2.2 Deepfake Detection Paradigms

Detection approaches can be organised along two axes: the temporal granularity of the analysis and the source of discriminative features. At the coarser level, intra-frame detectors classify individual images independently, whilst inter-frame detectors exploit temporal inconsistencies across video sequences. Within intra-frame detection, data-driven methods learn forgery features directly from image-level supervision, knowledge-driven methods encode domain-specific artefacts such as blending boundaries or resolution inconsistencies, and multi-stream methods combine both.

Among the most studied data-driven detectors, Xception repurposed the XceptionNet image classification backbone for binary forgery detection and became a widely adopted benchmark. A deliberately shallow architecture called MesoNet targets mesoscopic-level artefacts. The patch-forensics approach demonstrated that truncating standard networks at early convolutional blocks forces the model to focus on local, high-frequency forgery cues. The Multiple Attention framework framed deepfake detection as a fine-grained visual classification problem and proposed multi-scale attentional pooling to aggregate discriminative spatial evidence.

2.3 Benchmarking Literature

The most comprehensive deepfake detection benchmark to date was reported by Deng et al. (2024), who evaluated 13 detector variants across 7 datasets under a consistent protocol and with a purpose-built Imperceptible and Diverse (ID) test set. Under strictly controlled conditions, the large performance gaps reported in earlier non-uniform evaluations were substantially reduced. Our project is directly inspired by and based on the methodology of Deng et al. (2024), narrowing the scope to four architecturally representative detectors and adding efficiency metrics absent from the original benchmark.

3 Dataset and Experimental Setup

3.1 Dataset

The dataset consists of real and deepfake face images used for binary classification, with labels 0 (real) and 1 (deepfake). The held-out test partition contains 600 samples—150 real images and 450 fake images—representing a 1:3 real-to-fake class ratio. This imbalance is consequential for interpreting raw accuracy and F1 scores, and motivates the use of class-insensitive metrics such as Balanced Accuracy, ROC-AUC, and MCC. Face extraction was applied as a preprocessing step prior to training, ensuring each model input contains only the cropped and aligned face region, preventing background-driven spurious correlations.

3.2 Preprocessing and Augmentation

All images were resized to the input resolution required by each model (224×224 for Meso4 and Patch ResNet; 299×299 for Xception; 224×224 for Multiple Attention). Pixel values were normalised to the [0, 1] range and standardised using ImageNet mean and standard deviation values. Data augmentation during training included random horizontal flipping and random cropping to improve generalisation.

3.3 Implementation Details

All models were implemented in PyTorch 2.x and trained on CUDA-capable GPU hardware. The loss function used across all models was Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss), which applies a sigmoid activation internally and is numerically stable:

$$L = - [y \cdot \log \sigma(x) + (1 - y) \cdot \log(1 - \sigma(x))]$$

where x is the raw logit output, y is the binary label, and σ denotes the sigmoid function. The Adam optimiser was used with a base learning rate of 1×10^{-4} . A step-decay scheduler reduced the learning rate by a factor of 0.1 every 10 epochs. Training ran for 20 epochs with early stopping monitored on validation balanced accuracy.

4 Detection Architectures

4.1 Meso4

Meso4 is a compact convolutional neural network designed specifically for deepfake detection, operating at a “mesoscopic” level—a mid-frequency regime between raw pixel intensity and high-level semantic features. The intuition is that deepfake generation artefacts manifest as subtle, spatially localised inconsistencies in texture and colour statistics that do not require deep feature hierarchies to detect.

The architecture consists of four successive convolution-pooling blocks, each containing a 3×3 convolutional layer followed by batch normalisation, a ReLU activation, and a max-pooling layer. The resulting compact feature representation is flattened and passed through a single dense hidden layer with dropout before a sigmoid output. The entire network contains approximately 30,000 trainable parameters and requires only 0.06 GFLOPs per forward pass, making it by far the most computationally economical model in this study.

Meso4’s primary strength is its speed: exceeding 6,000 FPS on GPU, it is more than an order of magnitude faster than the second-fastest model. Its principal limitation is discriminative capacity—the shallow feature hierarchy cannot capture the complex, multi-scale forgery patterns that modern GAN-based manipulations exhibit.

4.2 Xception

Xception adapts the XceptionNet image classification backbone for deepfake detection by replacing the final classification head with a binary output. XceptionNet replaces standard convolutions with depthwise separable convolutions throughout, factorising the joint spatial-channel filtering operation into a depthwise spatial convolution followed by a pointwise channel projection. This factorisation reduces the number of multiply-add operations whilst maintaining expressive power, and is particularly well-suited to detecting subtle spatial inconsistencies that characterise many face manipulation methods.

The network processes 299×299 input images through entry, middle, and exit flow blocks, ultimately producing a 2048-dimensional global average-pooled feature vector before the classification layer. With 20.81 million parameters and 5.98 GFLOPs per image, Xception is the heaviest model in this evaluation (238.6 MB checkpoint, 331 FPS). Its strong pre-trained feature extraction capability allows it to identify subtle visual artefacts introduced by face swapping and reenactment, achieving the highest detection performance across all primary metrics.

4.3 Patch ResNet

Patch ResNet operates on local image patches rather than the full image, exploiting the observation that forgery artefacts in deepfakes are often spatially localised—appearing at blending boundaries, skin texture discontinuities, or in high-frequency regions that GAN generators struggle to reproduce faithfully. The architecture is derived by truncating a standard ResNet after its first residual block, retaining only the shallow convolutional layers and constraining the receptive field to a small local neighbourhood.

During training, patch-level binary labels are derived from the image-level label, and the loss is computed over the patch grid. At test time, patch-level predictions are averaged across the spatial grid to produce an image-level score. This aggregation provides implicit regularisation: a single anomalous patch is insufficient to trigger

a fake prediction, reducing false positive rates. Patch ResNet has 11.18 million parameters and requires 2.38 GFLOPs per image, running at 1,265 FPS with 0.79 ms latency.

4.4 Multiple Attention Detector

The Multiple Attention Detector frames deepfake detection as a fine-grained visual classification problem, where discriminative evidence is subtle and distributed across multiple spatial regions. The architecture introduces three key components: a multi-attention module generating several attention maps in parallel, each focusing on a different candidate forgery region; densely connected convolutional layers amplifying subtle texture artefacts in shallow feature representations; and bilinear attention pooling aggregating low-level textural and high-level semantic features guided by the learned attention maps.

An EfficientNet-B4 backbone performs feature extraction. A region-independent loss encourages diversity among attention maps, preventing collapse to a single dominant forgery region. The Attention-Guided Data Augmentation (AGDA) mechanism synthetically occludes attended regions during training, forcing the network to discover additional forgery evidence. The model contains 12.92 million parameters and requires 2.48 GFLOPs, achieving 1,208 FPS with 0.83 ms latency. It achieves a ROC-AUC of 0.841 and MCC of 0.493—very close to Xception’s 0.879 and 0.510—while requiring less than two-thirds the parameters and half the FLOPs.

5 Evaluation Methodology

5.1 Detection Performance Metrics

Let TP, FP, TN, and FN denote true positives (fakes correctly classified), false positives (real images misclassified as fake), true negatives (real images correctly classified), and false negatives (fakes misclassified as real), respectively. The deepfake class is treated as the positive class throughout.

Accuracy measures the proportion of all test samples correctly classified:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Accuracy is intuitive but misleading under class imbalance—a naïve classifier predicting all samples as fake would achieve 75% accuracy on our test set.

Balanced Accuracy is the average of sensitivity and specificity:

$$\text{Balanced Accuracy} = (TPR + TNR) / 2$$

where $TPR = TP / (TP + FN)$ and $TNR = TN / (TN + FP)$. A balanced accuracy of 50% corresponds to random performance regardless of class distribution.

Precision and **Recall** measure, respectively, the fraction of fake predictions that are correct and the fraction of actual fakes detected:

$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN)$$

Specificity measures the fraction of real images correctly identified as real: $TN / (TN + FP)$.

F1 Score is the harmonic mean of precision and recall:

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$$

Matthews Correlation Coefficient (MCC) accounts for all four cells of the confusion matrix and is robust to class imbalance:

$$MCC = (TP \cdot TN - FP \cdot FN) / \sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN))}$$

MCC ranges from -1 (perfect inverse) to $+1$ (perfect prediction), with 0 indicating chance performance.

ROC-AUC summarises the TPR-vs-FPR curve as the probability that the model scores a random positive example higher than a random negative. **PR-AUC** (average precision) summarises the precision-recall curve and is particularly informative under class imbalance.

5.2 Computational Efficiency Metrics

Computational efficiency is evaluated across five dimensions. FLOPs (Floating Point Operations) count the multiply-add operations per forward pass, providing a hardware-independent measure of model complexity. Parameter count quantifies learnable weights, determining memory consumption and checkpoint size. Model size in megabytes is the on-disk footprint. FPS measures real throughput on the evaluation GPU and is the most

practically relevant measure for batch processing. Latency in milliseconds per image captures single-sample inference time, which matters most for real-time applications.

6 Experimental Results

6.1 Training Behaviour

Figure 1 presents the training curves for all four models across 20 epochs, showing training loss (dashed), validation loss (solid), validation accuracy, and validation balanced accuracy.

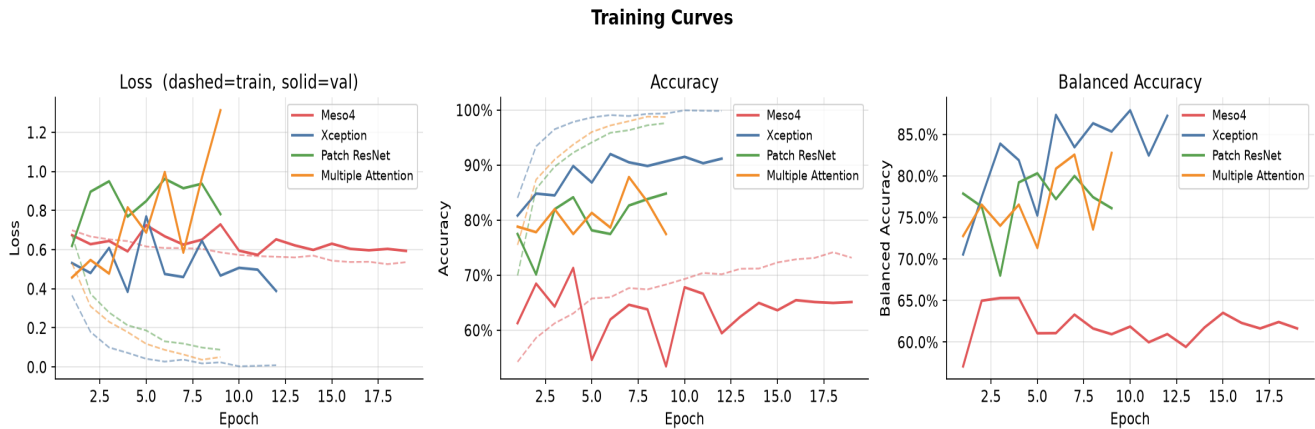


Figure 1. Training curves across 20 epochs. Dashed lines denote training values; solid lines denote validation values.

Xception exhibits the most stable and rapid convergence: its training loss decreases sharply in the first three epochs and the validation loss tracks closely, indicating minimal overfitting. Validation balanced accuracy climbs steadily to approximately 73–75% before stabilising.

Multiple Attention shows more oscillatory dynamics consistent with its multi-attention objective. Training loss spikes occasionally—most prominently around epoch 8—before recovering, suggesting the AGDA mechanism periodically challenges the model with difficult occlusions. Despite this instability, validation balanced accuracy converges to a level comparable to Xception.

Patch ResNet converges to a plateau by approximately epoch 10. Its balanced accuracy trajectory is moderately stable, achieving lower peak balanced accuracy than Xception and Multiple Attention, consistent with the patch-based approach’s more constrained representational capacity.

Meso4 shows the most distinctive pattern: slow loss decrease and validation balanced accuracy hovering near 60–65% throughout training. The near-zero MCC at test time (0.000) suggests the model collapsed to predicting almost exclusively the majority (fake) class, consistent with its 100% recall and 0% specificity in the confusion matrix.

6.2 Detection Performance

Table 1 summarises the complete set of detection performance metrics across all four models on the held-out test set.

Model	Accuracy	Bal. Acc.	Precision	Recall	Specificity	F1	MCC
Meso4	75.00%	50.00%	75.00%	100.00%	0.00%	85.71%	0.000
Xception	82.67%	73.56%	86.04%	91.78%	55.33%	88.82%	0.510
Patch ResNet	76.33%	69.33%	84.84%	83.33%	55.33%	84.08%	0.380
Multiple Attention	82.00%	72.89%	85.77%	91.11%	54.67%	88.36%	0.493

Table 1. Detection performance metrics on the held-out test set. Best values in each column appear in the Xception row.

Xception achieves the highest scores across all discriminative metrics: 82.67% accuracy, 73.56% balanced accuracy, 88.82% F1, 0.8791 ROC-AUC, 0.9531 PR-AUC, and 0.5100 MCC. The elevated ROC-AUC and PR-AUC confirm that Xception’s score distributions are well-calibrated and provide the sharpest real-versus-fake separation in the cohort.

Multiple Attention comes close on every metric: 82.00% accuracy, 72.89% balanced accuracy, 0.8411 ROC-AUC, 0.9361 PR-AUC, and 0.4925 MCC. The marginal gap from Xception—approximately 0.04 in ROC-AUC and 0.02 in MCC—is practically small and within the range reversible by hyperparameter tuning. Patch ResNet occupies third position with 76.33% accuracy and 0.7878 ROC-AUC; its balanced accuracy of 69.33% and MCC of 0.380 are substantively above random. Meso4’s 75% raw accuracy is driven almost entirely by predicting all samples as fake, as confirmed by its 50% balanced accuracy and 0.000 MCC.

6.3 ROC Analysis

Figure 2 displays the Receiver Operating Characteristic curves for all four models, illustrating the trade-off between True Positive Rate and False Positive Rate across all decision thresholds.

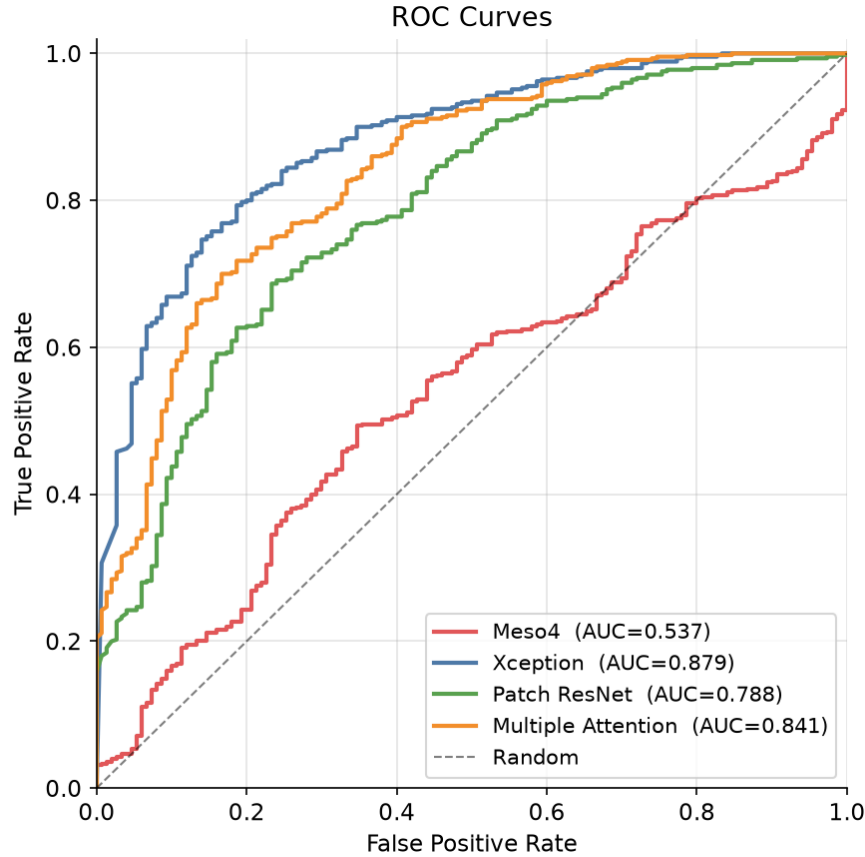


Figure 2. ROC curves. The dashed diagonal represents random performance (AUC = 0.5).

Xception's curve (AUC = 0.879) rises steeply from the origin, reaching a true positive rate above 0.75 while maintaining a false positive rate below 0.20—a region corresponding to deployable performance. Multiple Attention (AUC = 0.841) traces a closely parallel path, indicating broadly similar discriminative ability. Patch ResNet (AUC = 0.788) shows a shallower initial slope but maintains reasonable separability across the full range of thresholds. Meso4's curve (AUC = 0.537) barely rises above the random baseline, confirming that its score distribution does not meaningfully separate the two classes.

6.4 Precision-Recall Analysis

Figure 3 presents the Precision-Recall curves, which are particularly informative when the positive (fake) class is of primary interest and the dataset is imbalanced. The dashed horizontal baseline at 0.75 represents the precision achievable by a classifier predicting all samples as fake.

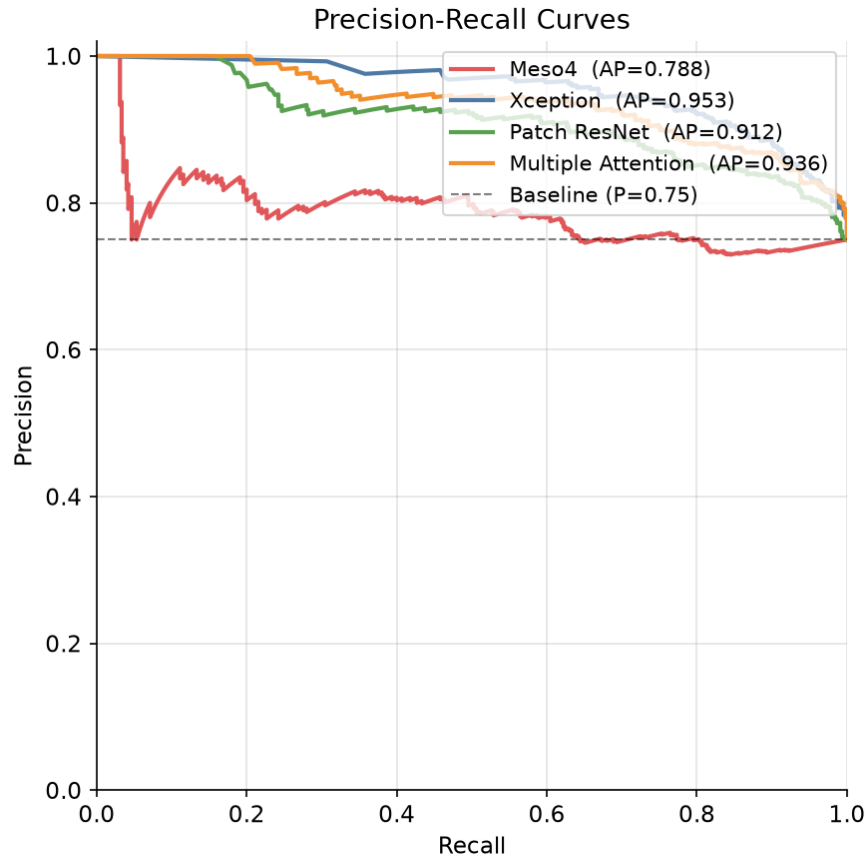


Figure 3. Precision-Recall curves. The dashed line at $P = 0.75$ indicates the trivial all-fake baseline.

Xception's curve ($AP = 0.953$) remains above 0.90 precision across nearly the entire recall range until recall approaches 0.85. Multiple Attention ($AP = 0.936$) closely tracks Xception's curve with slight divergence at high recall values. Patch ResNet ($AP = 0.912$) exhibits a steeper drop in precision at higher recall but still substantially outperforms the baseline. Meso4's curve ($AP = 0.788$) oscillates substantially in the low-recall regime and never maintains consistent high precision; its MCC of 0.000 is a more faithful summary of its discriminative value than its PR-AUC.

6.5 Confusion Matrix Analysis

Figure 4 presents the row-normalised confusion matrices for all four models, revealing qualitative patterns of classification errors.

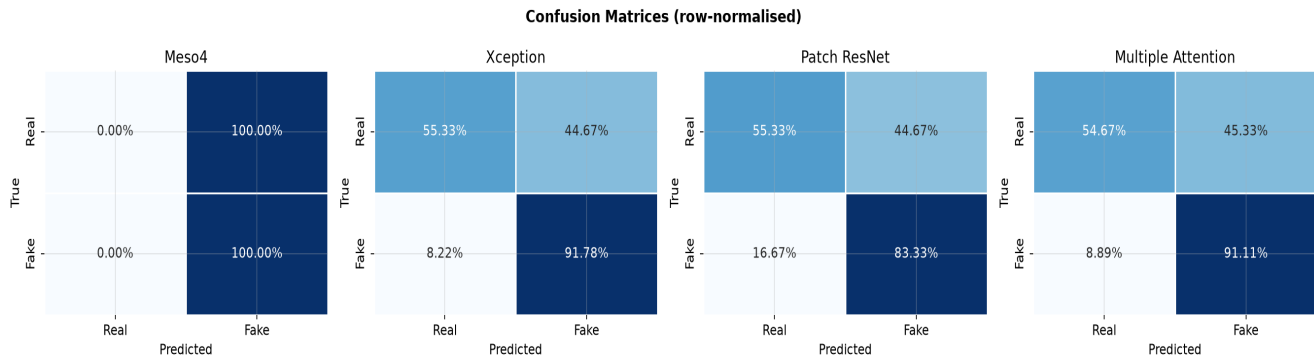


Figure 4. Row-normalised confusion matrices. Rows correspond to true labels; columns to predicted labels.

Meso4’s confusion matrix is the most diagnostic: every real image (100%) is misclassified as fake, and every fake image (100%) is correctly identified. This confirms that Meso4 collapsed to the majority-class prior under the default decision threshold of 0.5.

Xception correctly identifies 55.33% of real images and 91.78% of fake images. The false negative rate of 8.22% on fakes is the lowest among all models. The 44.67% false positive rate on real images reflects a defensible bias in security-critical applications where missing a deepfake is more costly than a false alarm.

Patch ResNet shows a 55.33% true negative rate on real images and 83.33% true positive rate on fakes. The higher false negative rate (16.67%) relative to Xception reflects the patch-based model’s tendency to require stronger or more widespread local evidence before committing to a fake classification. Multiple Attention achieves 54.67% specificity and 91.11% recall on fakes—very similar to Xception—confirming comparable decision characteristics at the default threshold.

6.6 Efficiency Analysis

Table 2 summarises the computational efficiency metrics for all four models.

Model	Params (M)	GFLOPs	Size (MB)	FPS	Latency (ms/img)
Meso4	0.03	0.06	0.3	6,143	0.16
Xception	20.81	5.98	238.6	331	3.02
Patch ResNet	11.18	2.38	128.0	1,265	0.79
Multiple Attention	12.92	2.48	148.0	1,208	0.83

Table 2. Computational efficiency metrics. FPS measured on a CUDA-capable GPU.

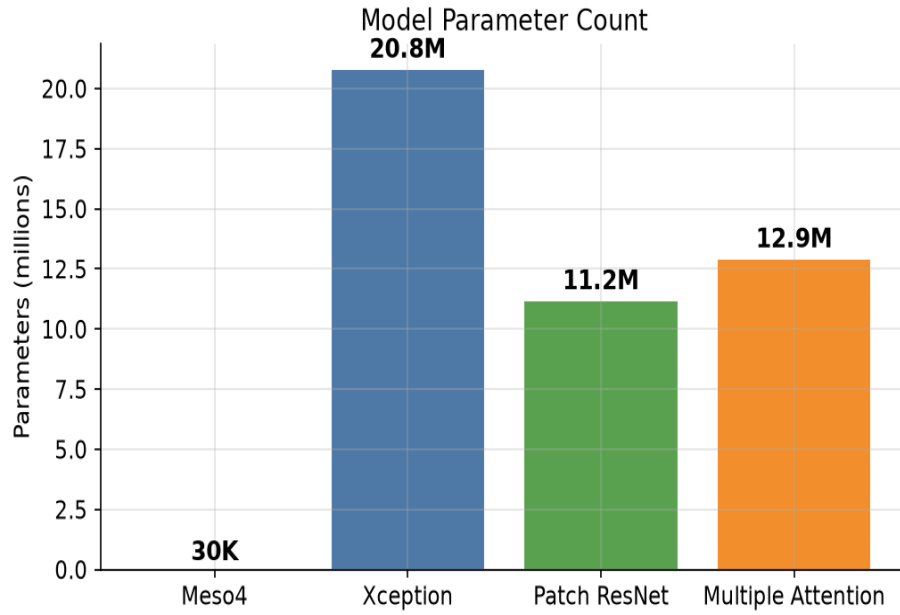


Figure 5. Model parameter counts in millions. Note the order-of-magnitude gap between Meso4 and the remaining models.

At 30,000 parameters, Meso4 is more than 370× smaller than Xception and approximately 370–430× smaller than Patch ResNet and Multiple Attention. The corresponding checkpoint sizes (0.3 MB versus 128–239 MB) reflect this difference directly.

Xception’s 5.98 GFLOPs represents the highest computational cost—approximately 2.5× greater than Patch ResNet and Multiple Attention—driven by the deeper network and larger spatial input. Its inference throughput of 331 FPS is sufficient for near-real-time single-stream video but may bottleneck high-throughput batch screening. Patch ResNet and Multiple Attention are closely matched on all efficiency metrics (2.38 vs 2.48 GFLOPs; 1,265 vs 1,208 FPS; 0.79 vs 0.83 ms latency), both running approximately 3.8× faster than Xception.

6.7 Accuracy-Efficiency Trade-off

Figure 6 plots balanced accuracy against FPS for all four models, capturing the accuracy-efficiency frontier.

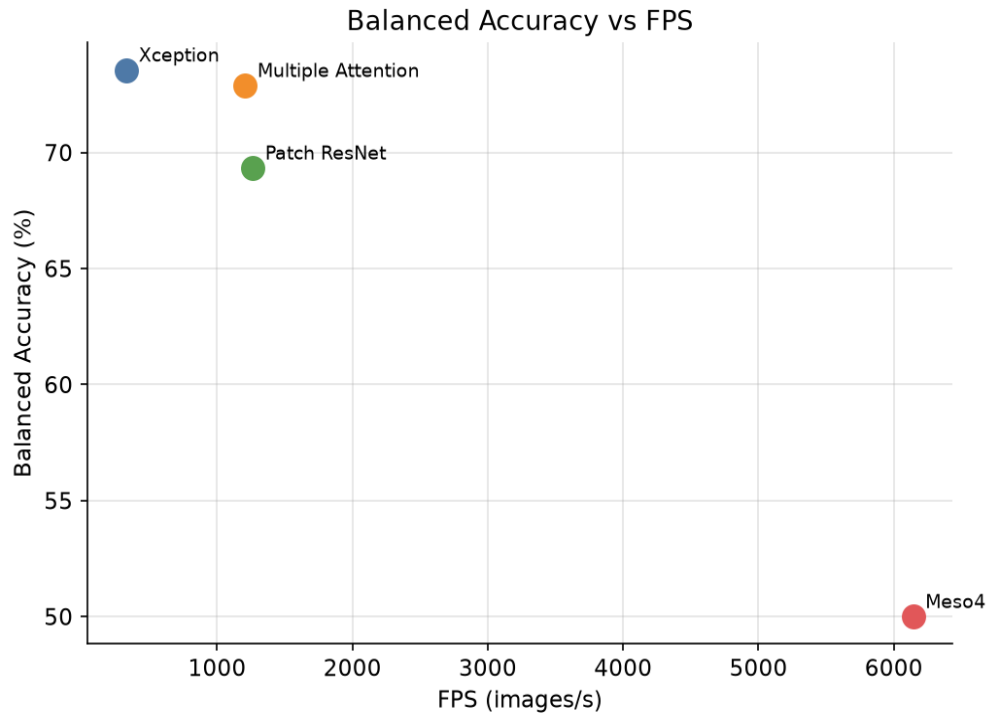


Figure 6. Balanced Accuracy (%) vs FPS. Points in the upper-left represent favourable trade-offs.

The plot reveals a clear bifurcation. Meso4 occupies the far right at exceptional speed but sacrifices all discriminative accuracy. Xception and Multiple Attention cluster in the upper-left, achieving high balanced accuracy at moderate speed. Patch ResNet occupies a middle position with slightly lower accuracy than both leaders but substantially higher throughput.

The trade-off frontier suggests that Multiple Attention is the Pareto-optimal choice: it approaches Xception's accuracy while running at nearly 4× the speed. For applications where maximum accuracy is paramount and throughput is not a constraint, Xception is the preferred choice. For resource-constrained environments such as mobile devices or edge hardware, Patch ResNet offers a practical compromise. Meso4 should be reserved for applications where raw detection throughput is the primary objective and discriminative balance is not critical.

7 Discussion

7.1 Why Xception Performs Best

Xception’s superior performance is attributable to the combination of a deep, expressive feature hierarchy—36 separable convolutional layers—and ImageNet pre-training. Depthwise separable convolutions enable the network to learn rich cross-channel feature combinations without the quadratic growth in parameters associated with standard convolutions, allowing a high-capacity model to remain tractable to train. The ImageNet pre-training provides strong low-level feature initialisation that transfers well to detecting subtle texture and colour artefacts. Xception’s 299×299 input resolution also preserves more high-frequency spatial detail than smaller inputs, which is important for detecting the fine-grained artefacts introduced by face manipulation.

7.2 Multiple Attention as Practical Choice

Multiple Attention’s strong performance relative to its parameter count reflects the efficiency of the EfficientNet-B4 backbone and the focused nature of attention-guided feature aggregation. By learning to attend to multiple candidate forgery regions simultaneously, the model allocates its representational capacity where it matters most. The bilinear attention pooling mechanism captures interactions between attention-weighted low-level and high-level representations that global average pooling cannot. Collectively, these design choices enable the model to match Xception’s discriminative capability at substantially lower computational cost, making it the most practically attractive choice when both accuracy and throughput matter.

7.3 Patch ResNet as an Efficient Compromise

Patch ResNet’s truncation at shallow network layers is motivated by the observation that deepfake artefacts are often localised in high-frequency texture regions visible to early convolutional layers but diluted as the feature representation becomes more abstract. Restricting the receptive field to small spatial neighbourhoods forces the model to make decisions based on local evidence alone, which tends to be less dataset-specific and more transferable across manipulation methods. The trade-off is reduced capacity to integrate global contextual cues—such as identity-level inconsistencies—that deeper architectures can exploit.

7.4 Meso4 and the Limits of Lightweight Design

Meso4’s failure to produce balanced predictions reflects a fundamental tension in lightweight deepfake detection. With only 30,000 parameters, the model lacks the representational capacity to model the diverse range of forgery artefacts in modern face manipulation datasets. Under a class-imbalanced training regime—or when the decision threshold is not optimised post-training—the model defaults to the majority class, yielding high raw accuracy but zero balanced accuracy and MCC. Threshold calibration or cost-sensitive training could recover some discriminative performance in practice.

7.5 Generalisation and Dataset Limitations

All results in this study are in-domain evaluations: models are trained and tested on splits from the same dataset. Generalisation to unseen manipulation methods, novel generative models, or heavily compressed media is likely to be substantially worse. As Deng et al. (2024) demonstrated in their comprehensive benchmark, deepfake detection models trained on one manipulation type transfer poorly to others, particularly when forgery artefacts are visually subtle or the post-processing pipeline differs. This limitation applies to all four models evaluated here.

The test set composition—150 real and 450 fake images—also introduces a potential source of bias. Under this class imbalance, raw accuracy is dominated by fake class performance, and models that default to the majority class are penalised only by balanced accuracy and MCC. Future work should consider equalised class ratios or explicit class-weighting in the loss function.

8 Conclusion

This report has presented a comparative benchmarking study of four deepfake detection architectures—Meso4, Xception, Patch ResNet, and Multiple Attention—evaluated under a unified training and evaluation pipeline. By controlling for preprocessing, dataset splits, and evaluation metrics, and following the methodology of Deng et al. (2024), the study provides a fair and interpretable comparison of the four methods.

Xception achieves the strongest detection performance across all primary metrics, including a ROC-AUC of 0.879 and an MCC of 0.510, at the cost of the highest computational overhead (20.81M parameters, 5.98 GFLOPs, 331 FPS). Multiple Attention provides near-equivalent detection capability—within 0.04 ROC-AUC and 0.02 MCC—while requiring substantially fewer resources (12.92M parameters, 2.48 GFLOPs, 1,208 FPS), making it the most practically attractive choice when both accuracy and throughput matter. Patch ResNet offers a balanced intermediate option with moderate accuracy (ROC-AUC 0.788) and high inference speed (1,265 FPS). Meso4, despite its exceptional speed (6,143 FPS), fails to produce balanced predictions under the default threshold, highlighting the discriminative limitations of extremely lightweight architectures.

These findings align with the qualitative conclusions of Deng et al. (2024): no single model dominates across all evaluation dimensions, and the choice of detector must be guided by the application’s specific requirements. Continued progress in this field will require both more powerful detection architectures and more rigorous, standardised evaluation protocols.

Reference

[1] J. Deng, C. Lin, P. Hu, C. Shen, Q. Wang, Q. Li, and Q. Li, “Towards benchmarking and evaluating deepfake detection,” *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 6, pp. 5112–5127, Nov./Dec. 2024.